ERG Surface Sampling ROVER Manual

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Last Revised: August 2026

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1 Introduction

The Remote Operated Vacuum for Environmental Science (ROVER) is designed to enable data collection on non-exhaust emissions over a wide sampling area. The ROVER uses a **centrifugal fan** to resuspend PM and two identical **gas diaphragm pumps** with two **PM2.5 cyclone separators** to collect simultaneously onto two **PTFE TE37 1µm diameter 25mm membrane filters**. In addition, it enables road surface imaging and the collection of telemetry data such as GPS and velocity through an attached **GoPro HERO7 Silver**, and particle concentration through an **AlphaSense OPC-N3**. It is supported by two **castor wheels** and four **all-terrain drive wheels** powered by an **18V DeWalt rechargeable battery**.

2 Component Layout and Naming Conventions

2.1 Section Convention

The ROVER can be separated into three main sections as follows:

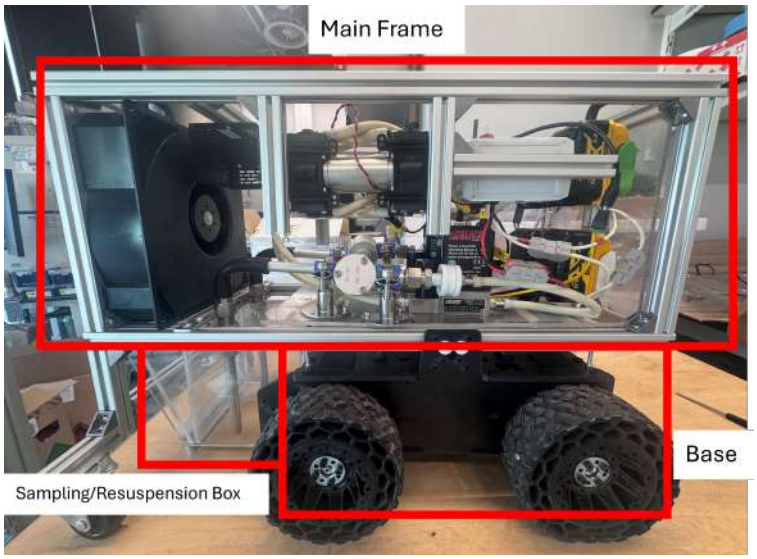


Figure 1: Side view of the ROVER with the major sections labelled

The base contains the two motors powered by an **18V DeWalt rechargeable battery** and is connected to the frame via seven connection points that will be detailed in Section 4.1. The main frame houses all of the environmental apparatus, and is connected to the sampling box.

In addition to the three sections on the main body, there is also a cover that slides down over the main frame, and a controller that is used to operate the ROVER remotely:



Figure 2: Side view of the ROVER cover.



Figure 3: Image of the ROVER controller.

2.2 Orientation

When in operation, the forward direction of the ROVER is in the direction of the sampling box, so that side will be defined as the front, and all other orientations are defined relative to the front.

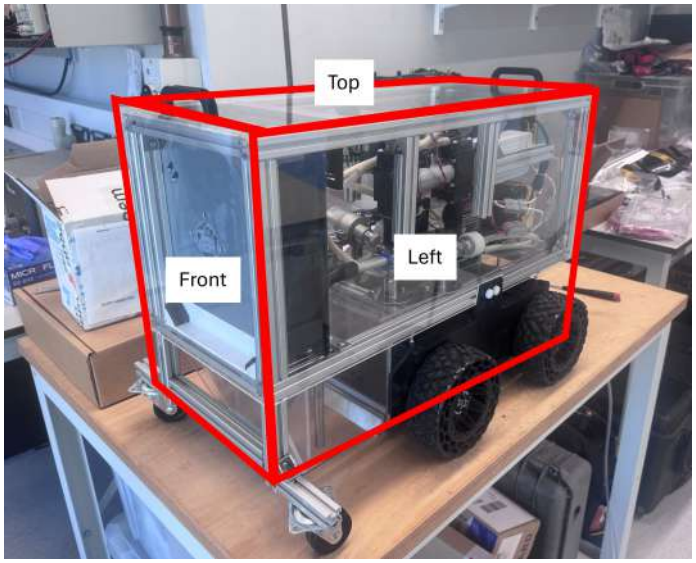


Figure 4: Perspective view of the ROVER with the front, left, and top sides labelled. The respective opposing sides, rear, right, and bottom are not labelled.

2.3 Subsystem Layouts

2.3.1 Physical/Mechanical Subsystem

This subsystem contains all the major hardware components of the ROVER. Their layout is illustrated in the following figures.

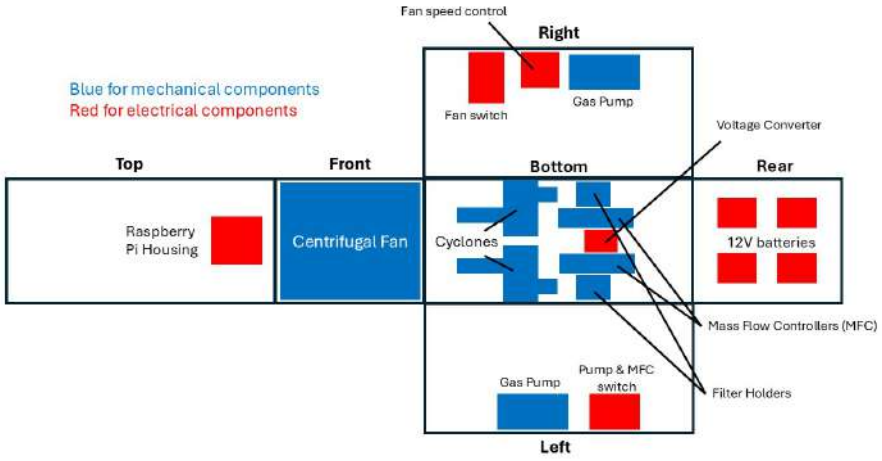
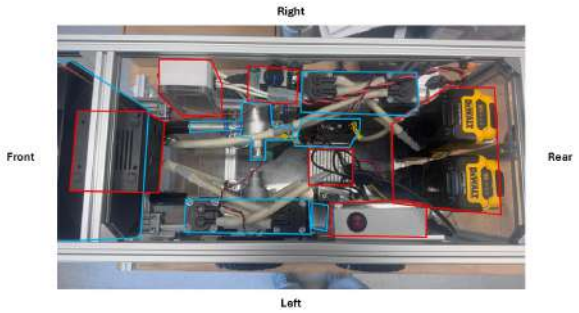
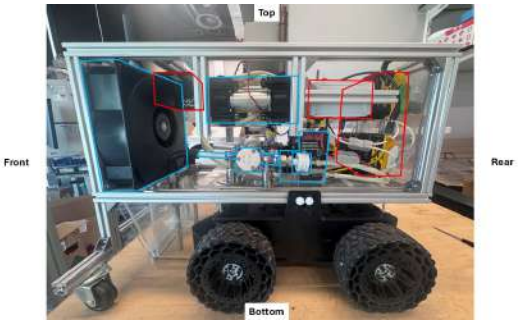


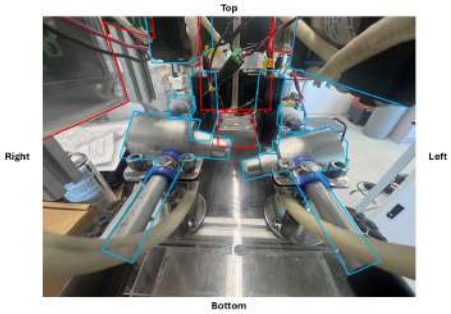
Figure 5: An unfolded view of the ROVER with the major components labelled. Mechanical components are labelled in blue, and electrical components are labelled in red.



(a) View from the top towards the bottom.



(b) View from the left towards the right.



(c) View from the front towards the rear.

Figure 6: Three different views of the ROVER, with major components outlined in corresponding colors.

2.3.2 Electrical Subsystem

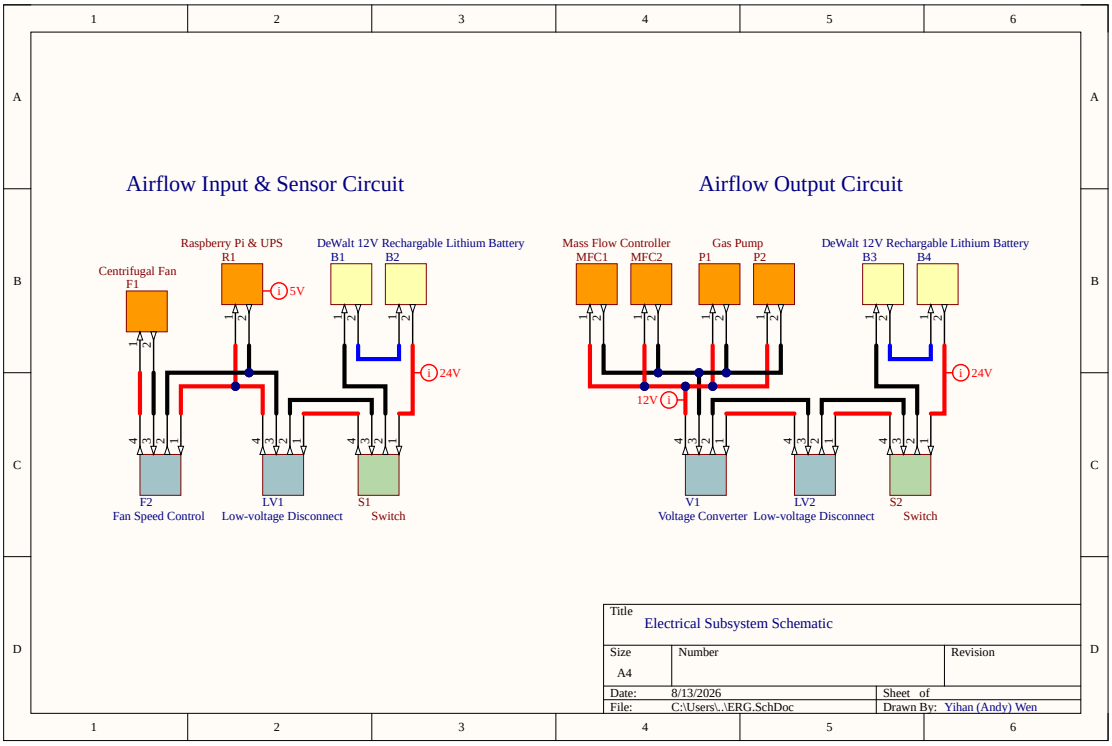
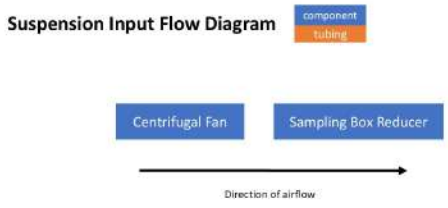
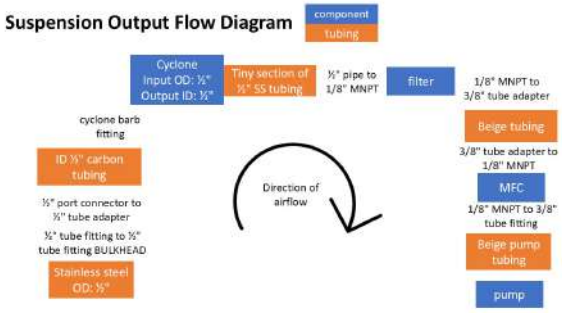


Figure 7: A schematic of the electrical subsystem.

2.3.3 Airflow Subsystem



(a) Sampling box input airflow diagram.



(b) Sampling box output airflow diagram.

Figure 8: Airflow diagrams for the sampling box, components are blue and tubing is orange, with Swagelok connectors annotated. Both output systems are identical.

3 Important Specifications

Type	Specification	Value
Dimension WxDxH (mm)	Main Frame Interior	560 x 200 x 220
	Main Frame Exterior	600 x 240 x 260
	ROVER (without GoPro)	635 x 405 x 430
	ROVER (with GoPro)	725 x 405 x 430
Weight (kg)	ROVER	**TO BE DETERMINED**
Battery (V)	Base Battery	18V DeWalt Lithium Battery
	Main Frame Batteries	12V DeWalt Lithium Battery
	Controller Batteries	1.5V AA Batteries
Flow Rate (L/min)		5
Filter Type		PTFE TE37 25mm diameter

Table 1: Important Specifications of the ROVER

4 Operation Instructions

In addition to covering the standard operation procedures for the ROVER, this section will also cover the process for detaching and attaching the main frame and the base, which is essential for maintenance and operation. Also, a section will be dedicated to adjusting the height of the castor wheels, which is important to maintain a balance between clearance and sample collection efficiency depending on the terrain and sampling conditions.

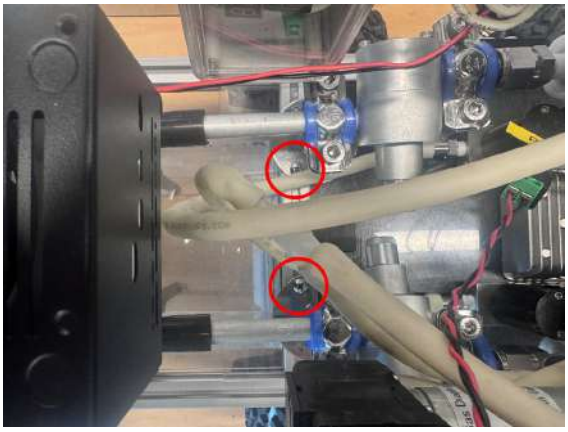
4.1 Detaching/Attaching the Main Frame and the Base

The procedure for detaching and attaching are very similar. To detach the main frame from the base, the following steps should be followed:

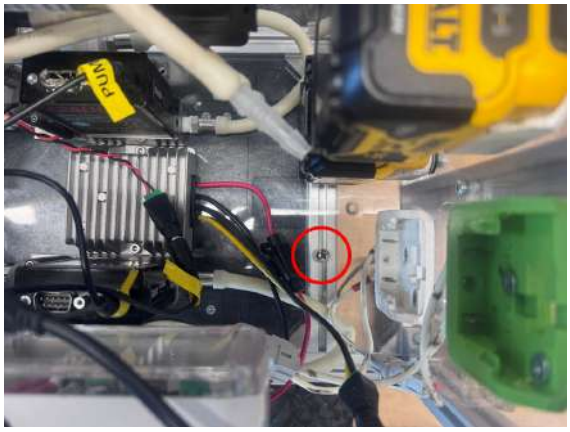
1. Remove the cover from the main frame. Note that the grooves on the side of the cover as shown in Figure 2 should be aligned with the grooves on the side of the base, located towards the rear of the ROVER. It is important to keep this in mind when reattaching the cover to the main frame.

2. If the 12V DeWalt batteries are installed, remove them from the main frame to access the connection points.

3. Unscrew the two screws towards the front of the base side that also double as mounting screws for the cyclone stands, as shown in Figure 9a. In addition, unscrew the screw towards the rear of the base side that is positioned underneath the 12V battery that was removed, as shown in Figure 9b.



(a) Front screws to be removed.



(b) Rear screw to be removed.

Figure 9: Screws located in the main frame to be removed in order to detach the main frame from the base.

4. Unscrew the four screws located in pairs on either side of the main frame. One such pair is shown in Figure 10. The other pair is located directly opposite of the main frame.

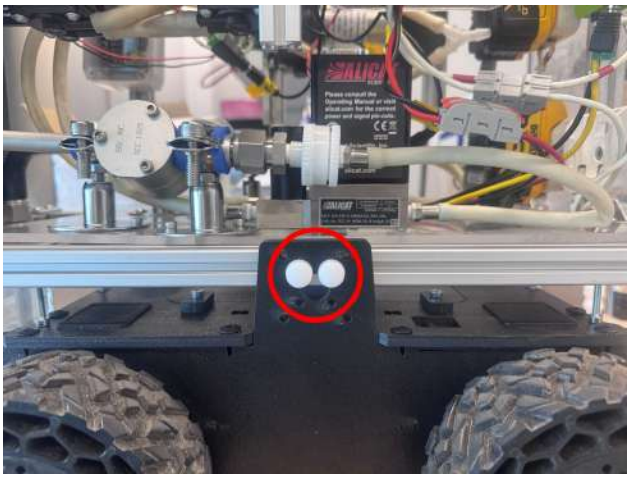


Figure 10: Side screws to be removed.

5. Carefully lift the main frame off the base.

You will now have access to the base and the battery compartment, which will be revisited in Section 4.3. To attach the main frame back onto the base, simply reverse the steps above. In addition to noting the orientation of the cover, it is also important to ensure the main frame is oriented correctly with respect to the base. The sampling box should be positioned towards the front of the base, between the two tabs circled in Figure 11.

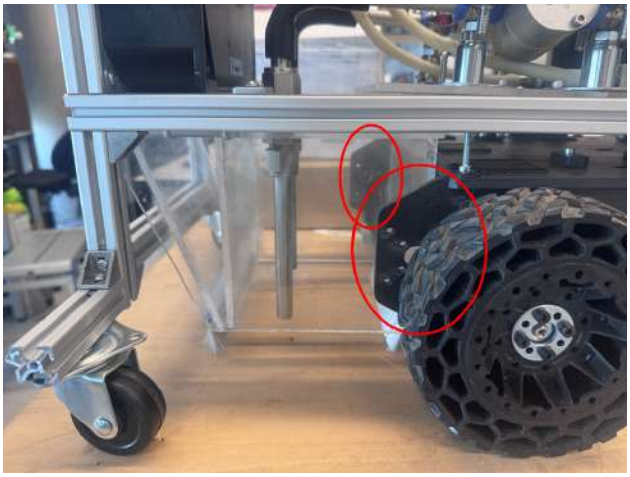


Figure 11: Tabs on the base that the main frame should be positioned between when reattaching.

4.2 Adjusting the Height of the Castor Wheels

To achieve optimal efficiency for sample collection, it is best for the sampling box to be as close to the ground as possible. However, often the sampling environment requires a certain amount of clearance. To adapt to different sampling environments, the height of the castor wheels can be adjusted to achieve a balance between clearance and sample collection efficiency.

To adjust the height of the castor wheels, simply unscrew the screw circled in Figure 12 and adjust the height of the castor wheel to the desired height. Then, screw the screw back in to secure the castor wheel in place. Do so for both sides of the ROVER to ensure that the main frame is level.



Figure 12: Castor wheel adjustment screw.

4.3 Pre-Operation

It is important to conduct a pre-operation check to ensure that the ROVER is in proper working condition before operation. The following steps should be followed:

1. If the main frame is attached to the base, detach it according to the procedure in Section 4.1.

2. Open the battery compartment by rotating the latches in Figure 13a to check the battery level of the 18V DeWalt battery. If the battery is low, remove it to charge before replacing it.



(a) Closed compartment with latches circled.



(b) Open battery compartment.

Figure 13: Images of the battery compartment and the 18V DeWalt battery.

3. Ensure that the switch on the 18V battery dock is switched off as shown in Figure 14a. This switch is redundant and forms an OR gate with the exterior switch on the base in Figure 14b. This means that if the interior switch is on, the exterior switch will not be able to correctly shutoff the motors, leading to excess battery drain.



(a) Interior switch on the 18V battery dock.



(b) Exterior switch on the base.

Figure 14: Images of the the interior and exterior switches on the base, both in the off position.

4. Close the battery compartment and reattach the main frame to the base according to the procedure in Section 4.1.

5. Verify that the filter holders contain clean filters by removing the attached metal tubing from the cyclone separators and unscrewing the filter holders as shown in Figure 15.

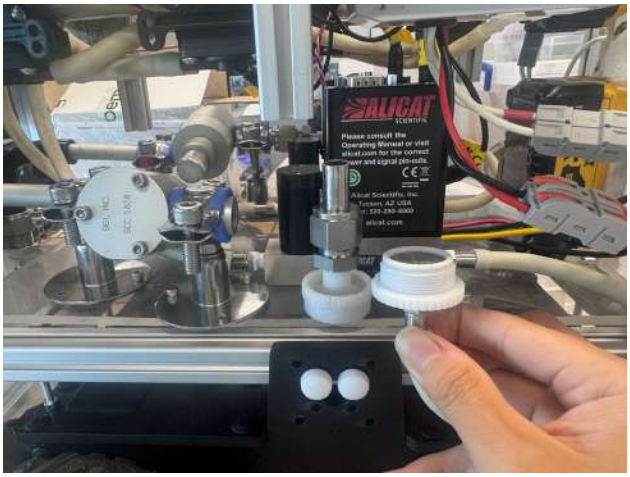


Figure 15: Example of the filter holder separated from the cyclone and unscrewed. The filter shown is dirty and should be replaced with a clean filter.

6. Unscrew the cyclone separator collection chambers shown in Figure 16 to ensure that they are empty, cleaning if needed. It may be easier to remove the cyclone separators from their supports entirely to access the collection chambers.

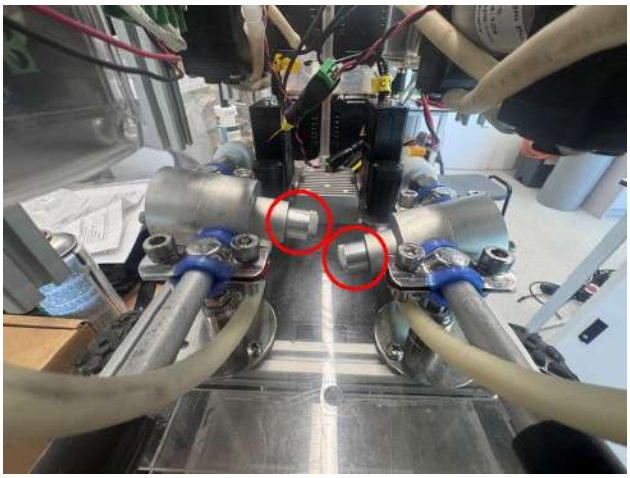


Figure 16: Cyclone separators with the collection chambers circled.

7. Check that the 12V DeWalt batteries are charged and installed in the docks at the rear of the main frame.

8. Place the cover back over the main frame, again ensuring that the grooves on the cover are aligned with the grooves on the base.

9. Verify that the GoPro is charged and has sufficient storage space. If it is not attached to the cover, position it as shown in Figure 17 and attach it. It may be necessary to wipe down the suction cup and the cover to ensure secure attachment.



Figure 17: Example of the GoPro attached to the cover.

10. Ensure that the controller has sufficient battery power.

4.4 Mid-Operation

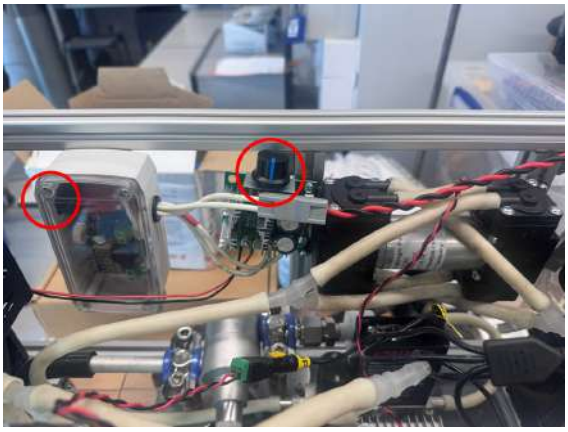
4.4.1 Driving the ROVER

There are a couple very important points of consideration before driving the ROVER. First, the ROVER is not designed to be driven on roads with concurrent traffic. Second, it is important to adjust the clearance (see Section 4.2) depending on the sampling conditions. Third, the motors are very sensitive to controller input, so take care in controlling to avoid sudden accleration or deceleration. When pausing movement, remember to decelerate slowly and always turn either the controller or the motors off to avoid any injury and damage to the ROVER. To drive the ROVER, follow the steps below:

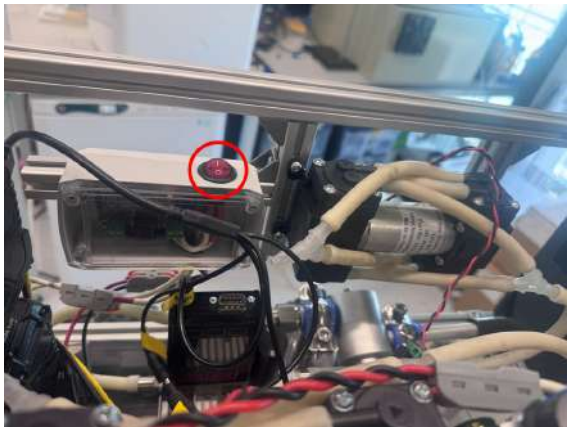
1. Turn on the controller.
2. Turn on the exterior switch on the base. When the controller successfully pairs, you should hear a series of beeps.
3. Move the left joystick left and right to turn. Move the right joystick up and down to move forwards and reverse. Once again, the forward direction should correspond to the direction of the sampling box.
4. Remember to decelerate slowly when stopping, and turn off the controller or the motors when pausing movement.
5. When finished, turn off the exterior switch on the base and then turn off the controller.

4.4.2 Collecting Samples

Sample collection is done using the centrifugal fan and the two identical gas pump and MFC systems. The air input (fan) and output (pumps) have separate switches, located on either side of the main frame, as shown in Figure 18.



(a) Controls for the centrifugal fan. The switch is circled but hidden from this angle. There is also a rotary knob to control the fan speed.



(b) Switch for the gas pumps and the MFCs.

Figure 18: Image of all switches and controls for sample collection.

To differentiate between the switches, one way is to remember that the fan switch has an additional speed control knob located next to it. Alternatively, simply remember that the fan switch is positioned closer to the fan towards the front of the ROVER.

4.4.3 Collecting Video and Telemetry Data

The GoPro is fairly intuitive to use. One thing to note is that the telemetry data is dependent on signal strength and satellite availability, so better results will be achieved in more open areas.

4.5 Post-Operation

Below is a procedure for post-operation maintenance of the ROVER:

1. Ensure that the exterior switch on the base and the controller are turned off.
2. Remove the GoPro to download and process graphic and telemetry data. Ensure that the GoPro is charged and has sufficient storage space for future use.
3. Remove the cover from the main frame, wipe or clean it if necessary.
4. Remove the 12V DeWalt batteries from the main frame and charge them if needed.
5. Carefully detach the filter holders from the cyclone separators (refer to Figure 15) and place the samples in a petri dish for analysis.
6. Unscrew the cyclone separator collection chambers and empty them. Again, it may be easier to remove the cyclone separators from their supports entirely.
7. Detach the main frame from the base according to the procedure in Section 4.1.
8. Open the battery compartment and remove the 18V DeWalt battery from the base, charging it if needed.

5 Future Considerations

There are a couple important improvements or alterations that need to be done in the future.

- Finalize the name. Options include:
  - Roadway Observation and Collection Kinetic Instrument (ROCKI)
  - Dynamic Aerosol Vacuum Equipment (DAVE)
  - Aerosol Non-Exhaust Data Instrument (ANDI)
- Housing for the fan speed controller.
- Weighing of the ROVER to complete Table 1.
- The full functions of the OPC and the raspberry pi programs need to be verified and tested. This manual needs to be revised to include procedures for their usage.
- Testing needs to be done on a number of parameters, including:
  - The optimal fan speed for resuspension.
  - The optimal height of sampling in the sampling box
  - Concentration of particles collected at different speeds and heights
  - Duration of sampling
- Fabrication of more robust sampling box and cover components.
- Procedures to clean the sampling box and the brushes located at the bottom of the sampling box to prevent cross-contamination between samples.

6 Index for Major Components

Below is a reference table for links to all the important components:

Type	**Component and Hyperlink**
Frame	ValuFrame Series 5 Aluminium Profiles
	Clear Perspex for Cover and Sampling Box
Base	Main Base Body
	Controller
Sampling Components	Centrifugal Fan
	Gas Pump
	Mass Flow Controller
	Cyclone Separator
	Filter Holder
	Filters
Other Components	Raspberry Pi Model 5 16GB RAM
	Pi Board UPS
	AlphaSense OPC-N3
	GoPro HERO7 Silver

Table 2: Links for major components